CYBR-2200 Network Analysis
Week 7 Final Project
Network Traffic Analysis Report

Purpose:
This assignment is the final analytical project for this course. You will synthesize the skills developed throughout the term to produce a professional network traffic analysis report. The goal is to demonstrate your ability to interpret network data, identify security-relevant activity, and communicate findings clearly.

This report represents portfolio-quality work.

Project Overview

You will analyze the packet capture files provided for Week 7 and produce a formal written report. Your analysis should be evidence-based, clearly reasoned, and written in a professional tone.

Report Requirements

Your report must include the following sections. Use clear headings for each section.

1. Executive Summary

Briefly summarize what was observed in the network traffic.
• What happened?
• Why it matters?
• What should be done next?
(1–2 paragraphs)

2022-MTA-workshop-exercise-1-of-5

This PCAP is a typical representation of an active enterprise workstation seen in an office setting and Windows domain. TCP accounted for 86.1% of observed packets. TLS encrypted communications accounted for 58.1%, over half of the total bytes used on the system. Active Directory Operations were observed through the typical internal-domain protocols such as SMB2, LDAP, Kerberos, and DCE/RPC. HTTP analysis revealed a single ZIP file download, but no other anomalies such as abnormal behavior, communications, or command-and-control patterns were seen. This behavior is consistent with normal enterprise workstation activity.

2022-MTA-workshop-exercise-2-of-5

This PCAP appears to be a typical demonstration of concentrated encrypted communication between an internal host at 10.5.31.139 and a single external IP address at 64.44.101.193. After conducting statistical and observational analysis, TCP traffic represented 95% of all total packets captured with TLS representing 78% of

the total volume of bytes captured. In contrast to PCAP 1-of-5, this PCAP demonstrates sustained and repeated communication via TLS to a specific external host with over 4935 packets alone transmitted between the internal host and external host. The sessions were observed to be prolonged and coupled with the high packet count, this shows a potential command-and-control behavior based on sustained encrypted communication rather than typical-user browsing behavior.

2022-MTA-workshop-webserver-scanning-1-of-2

After statistical analysis of the PCAP, the capture exhibits large-scale web server scanning activity with the IP 188.165.143.159 as the target. Analysis disclosed an advanced level of ICMP traffic and multiple, short-lived TCP connections to port 80 from multiple external IP addresses. Automated probing of common active directories such as phpMyAdmin and other web applications were demonstrated through multiple HTTP requests. The traffic patterns are consistent with reconnaissance and vulnerability scanning rather than typically seen web browsing behaviors.

2022-MTA-workshop-webserver-scanning-2-of-2.pcap

After statistical analysis of the packet capture, large-scale distributed web scanning activity was demonstrated against the target 54.191.67.23. The majority of the traffic comprises short-lived TCP sessions to port 80 from a multitude of external IP addresses. This is a traffic pattern consistent with automated reconnaissance behavior instead of legitimate workstation activity. No successful exploitation or exfiltration of data is examined in this PCAP.

2. Network Environment Overview

Describe the network environment as observed in the traffic.

- Internal vs external traffic
- Types of hosts and devices
- General traffic patterns

2022-MTA-workshop-exercise-1-of-5

As a typical small office network, within the subnet, this specific network operates on 10.5.31.0/24. The most active internal host observed was 10.5.31.182, which created the most traffic via packet count. This would corroborate a Windows workstation based on traffic patterns. The server and gateway were observed to be most likely 10.5.31.5 and 10.5.31.1. Communication externally to the system was mostly over TCP port 443, showing HTTPS encrypted communication, and port 80, showing unencrypted HTTP communication. SMB2, LDAP, Kerberos, and DCE/RPC were observed over the PCAP, showing an Active Directory domain. DNS and ARP traffic were observed with no anomalies.

2022-MTA-workshop-exercise-2-of-5

The observed internal network operates within the 10.5.31.0/24 subnet. The primary workstation that appears to be generating the traffic is 10.5.31.139. Other gateway and additional systems on the network include IPs 10.5.31.1 and 10.5.31.5. HTTPS over port 443 was the primary mode of encrypted communication external to the network. Domain-based operations were present with the protocols SMB2, LDAP, Kerberos, and DCE/RPC. However, the difference between this and exercise one, is this is traffic is heavily concentrated between a single external IP address 64.44.101.193 and the internal workstation 10.5.31.139.

2022-MTA-workshop-webserver-scanning-1-of-2

The PCAP analysis determined that the server identified is 188.165.143.159. This server received over 51,000 packets throughout the timeframe of the PCAP. Most of the traffic originated from external IP sources, each creating a small number of packets (>4 packets) directed to port 80. The elevated ICMP packet traffic could be an indication of host discovery activity before the web application probing. Unlike the previous two PCAPs, this packet capture focuses on a server being targeted by multiple external sources rather than focusing on a single internal workstation that would initiate the outbound traffic.

2022-MTA-workshop-webserver-scanning-2-of-2.pcap

Conversation and Endpoint analysis reveals the public-facing server's IP at 54.191.67.23. Traffic to the server observed originates from a multitude of unique external IP addresses. Each address initiates a small TCP session to port 80. However, unlike the previous PCAPs that involved typical workstation behavior or sustained session and encrypted communication, this PCAP focuses on a single web server and the inbound connections targeting that server. The network traffic behavior is indicative of an internet-exposed server, the recipient of uninvited active and passive reconnaissance traffic.

3. Baseline Traffic Analysis

Explain what normal traffic looks like in the capture.

- Common protocols
- Typical communication patterns
- Expected behavior

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Protocol Hierarchy was studied and determined that TCP accounted for 86.1% of all packets in the PCAP. TLS was also accounted for 58.1% of all bytes. This indicates that a majority of the packets were encrypted, consistent with modern web browsing. 11% of the traffic consisted of SMB2, typical of internal-file sharing and domain communication. LDAP and Kerberos were also observed which further

corroborates a typical Windows domain. DNS traffic was also consistent with typical traffic patterns with no abnormal name resolution behavior. After analyzing the conversations, the workstation 10.5.31.182 communicated with a typical range of external IPv4 addresses of short-lived sessions, also consistent with normal web browsing behavior. This type of web browsing did not show signs of automated beaconing.

2022-MTA-workshop-exercise-2-of-5

Protocol Hierarchy statistics reveal that 95% of all packets were TCP traffic, with 78% of the total byte volume accounting for TLS traffic. This shows that the majority of traffic across the network was encrypted. While encrypted traffic is typical on a network, conversation and endpoint analysis showed that the external network 64.44.101.193 connected to the internal workstation 10.5.31.139 a disproportionate number of times with the session duration lasting upwards of 600 seconds, using thousands of packets more than any other conversation between the internal 10.5.31.139 and any other external IP. This capture exhibits disproportionately high, sustained, encrypted communication between the two hosts.

2022-MTA-workshop-webserver-scanning-1-of-2

Protocol Hierarchy statistical analysis shows that TCP accounts for approximately 70% of the total packet amount with ICMP traffic comprising nearly 25% of the total. The high percentage of ICMP traffic can indicate an active network probe instead of standard server traffic. TCP conversation analysis reveals repeated, short-lived sessions to port 80 with very small byte transfers (less than 220 bytes per session). These characteristics deviate from typical browsing activity seen in PCAPs, which would be seen with longer sessions and more distinct content.

2022-MTA-workshop-webserver-scanning-2-of-2.pcap

Protocol Hierarchy statistical analysis shows the TCP protocol accounts for approximately 90% of the total packets, with a much lower amount of ICMP traffic (6.5%) compared to the previous PCAP. The TCP traffic conversations displays thousands of short-sessions specifically directed at port 80, each session being small with a very small byte amount. This is indicative of a pattern deviation from typical browser behavior, which would have included longer sessions and data transfers to multiple external hosts. This PCAP showed none of those behaviors.

4. Identified Anomalies or Malicious Activity

Describe any suspicious or malicious activity observed.

- What traffic deviates from the baseline?
- Why is it concerning?
- Reference specific indicators from the traffic

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One traffic deviation observed in the PCAP was an HTTP GET request from the IP 10.5.31.182 to the external IP 216.172.172.187 which included ZIP archive file F121235223.zip. While Zip files can be potential payload staging, this was determined to be consistent with typical use that posed no threat. This was supported by the absence of new malicious behaviors, no repeated downloads, persistent external communications, and no peculiar follow-up communications were observed after download. In addition, there were no TCP sessions of unusual length, single isolated or repeated connections to an external IP indicative of malware. All other traffic patterns observed were consistent with a typical enterprise Windows office workstation.

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Conversation analysis conducted displays that 10.5.31.139 communicated a disproportionately high amount with the external IP 64.44.101.193 over TCP port 443. A TLS handshake was followed by the encrypted communication between the two hosts. This concentration of packet data, length of sessions, and consistent encryptions to the single external IP host deviates from typical traffic patterns. This pattern is consistent with a command-and-control communication or automated beaconing such as seen from the Quakbot.

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After reviewing the conversation statistical analysis, the PCAP shows multiple external IP addresses starting TCP SYN requests to the IP address 188.165.143.159 on port 80. Each session was brief, consisting of approximately four packets, followed by an HTTP GET request which targets the most common administrative or vulnerable web application directories such as /phpMyAdmin/, /admin/, and other similar paths. The server constantly responds with HTTP 404 Not Found responses, which indicates a pattern of automated web application scanning and enumeration instead of legitimate attempts to access the server.

2022-MTA-workshop-webserver-scanning-2-of-2.pcap

Conversational analysis of the PCAP exposes multiple SYN packets and short TCP handshakes from an abundant of external IP addresses all targeting 54.191.67.23 on port 80. The sessions are very short, contain very small payloads by byte, and show no authentication. This is consistent with behaviors seen during automated port scanning or active and passive reconnaissance. The large-scale scanning of the target using a multitude of host IPs suggest this could be a targeted attack rather than isolated probing.

5. Evidence and Indicators

List and explain key indicators of compromise or concern.
Examples include:
• IP addresses and ports
• Domain names or URLs
• Protocol misuse patterns

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Evidence and Indicators of compromise were negligible as key observations utilizing statistical and observational analysis showed standard operating procedures for a Windows office system. The most active internal host observed was 10.5.31.182. The internal subnet was 10.5.31.0/24. The dominant protocol utilized throughout the PCAP was TCP at 86.1% of all packets analyzed. Encrypted traffic accounted for 58.1% of all bytes throughout the system through the TLS protocol. The SMB2, LDAP, Kerberos and DCE/RPC were all observed showing Active Directory protocols in place and being used. Anomaly consisted of HTTP Zip file download from 216.172.172.187 but was determined to be non-threatening. No repeated or long-duration sessions were observed. No repeated anomalous DNS queries were observed. No concentrated communication or data transfer to a single external IP was observed. No uncommon TCP or UDP traffic was observed. This was consistent with a typical Windows office workstation.

2022-MTA-workshop-exercise-2-of-5

Evidence and indicators of compromise suggest an abnormal traffic pattern and should be further investigated. After statistical analysis of the PCAP, key indicators observed include the most active internal host was 10.5.31.139. The internal subnet was 10.5.31.0/24. The dominant protocol utilized was TCP accounting for 95% of total packets. TLS protocol accounted for 78% of the total byte volume. The dominant external host IP communicated with was 64.44.101.193. There was a disproportionate amount of traffic between the internal host (10.5.31.139) and the external host (64.44.101.193). Approximately 4,945 packets were exchanged between the internal and external host with session intervals lasting upwards of 601 seconds. Repeated TLS handshakes and encrypted data exchange in excess of 2MB were also observed between the two hosts. These indicators and abnormalities could potentially indicate a compromised system. Further investigation is necessary.

2022-MTA-workshop-webserver-scanning-1-of-2

Evidence and indicators of compromise indicate reconnaissance behavior against the targeted server 188.165.143.159. There was evidence of increased ICMP traffic (24.5%) along with TCP port 80 scanning activity. Numerous distinct external IP addresses are the source of the repeated, short-lived TCP sessions. The HTTP Get requests always followed the TCP activity, targeting the administrative directories

with constant 404 Not Found responses being transmitted back. There were no large data transfers or authenticated sessions within the PCAP. These indicators are a demonstration of reconnaissance behavior against a server.

2022-MTA-workshop-webserver-scanning-2-of-2.pcap

Evidence and indicators of compromise strongly support active and passive reconnaissance scanning activities against the target server 54.191.67.23. TCP was the dominant protocol used, comprising up to 90% of all packets. There were numerous, unique external IP addresses using thousands of short-lived TCP sessions which also allowed repeated scanning to port 80. With these small byte transfers, TCP SYN handshakes, and no sustained HTTP sessions such as what is seen in typical behaviors indicate there is supportive evidence of scanning activity to the target server.

6. Analyst Assessment and Recommendations

Provide your professional assessment.

- Is this a confirmed incident, suspicious activity, or policy violation?
- What actions would you recommend?
- What additional data would help confirm your assessment?

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After analysis of the Protocol Hierarchy, Endpoints, and Conversations, this PCAP does not appear to demonstrate any symptoms typically associated with malware within a network. This was observed through typical encrypted HTTPS traffic and domain-related protocols such as SMB2, LDAP, and Kerberos. While a ZIP file was observed and analyzed being downloaded over HTTP, no following symptoms of malware or indicators of compromise were discovered. Currently, this PCAP appears to be typical network behavior and harmless. No immediate containment or remediation actions are recommended or required at this time.

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After statistical analysis of the Protocol Hierarchy, Conversations, and Endpoints as well as packet data, this capture strongly suggests command-and-control style communication behavior consistent with malware infection such as what was observed in the Quakbot PCAP. The long duration of sessions between the internal (10.5.31.139) and external (64.44.101.193) hosts, number of packets 4945 (3000 more than any other IP conversation), combined with the concentration of network traffic to the single host strongly indicates potential compromise consistent with command-and-control behavior. This differs from the first PCAP as the first provides standard baseline pattens, this behavior is highly indicative of command-and-control communication and strongly suggests potential compromise. Immediate action is recommended and necessary to keep the other internal network components safe to include isolation of the infected workstation, conducting

endpoint forensic analysist to determine its attack vector to the workstation, reviewing DNS logs, and blocking all communication to the external IP until further investigations are conducted.

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After analysis of the Protocol Hierarchy, Conversations, Endpoints, TCP/UDP and HTTP traffic, the PCAP represents a web server reconnaissance and directory enumeration example. The degree and model of probing indicates the automated vulnerability scanning while no successful exploitation seen in the PCAP. Recommended actions include the implementation of intrusion prevention and detection alert system, rate limiting, reviewing the firewall logs, enabling firewall protections for web applications, and network monitoring for more attempts at exploitation of the server. Reviewing the server of all exposed directories, administrative interfaces, and unnecessary public services should be conducted through an IT audit.

2022-MTA-workshop-webserver-scanning-2-of-2.pcap

After statistical analysis of the Protocol Hierarchy, Conversations, and Endpoints, the behavior analysis through the PCAP appeared to represent distributed web server reconnaissance. While no successful attempts to compromise the target server were captured, the external-facing web server is actively being scanned and probed. Recommended actions to minimize the ability for the server to be compromised include enabling rate limiting, immediately deploying a web application firewall if not already complete, reviewing firewall logs, monitoring for repeated source IPs to find the culprit, and ensuring that any unnecessary services are not exposed to the public needlessly. Continued monitoring is advised to detect and interrupt any further potential follow-up exploitation attempts.

Formatting and Submission Requirements

- Length: 3–5 pages
- Format: DOCX or PDF
- Professional tone and complete sentences required
- Do NOT include PCAP files or raw packet dumps
- Screenshots should be minimal and only used when necessary

Submission Instructions

- Upload your report to Blackboard under:
  Week 7 Final Network Traffic Analysis Report
- Upload the final version of this report to your GitHub portfolio repository
- Blackboard is the official grading submission